

PRACTICE REPORT

COMPUTER NETWORKS

Lab 3

COMPUTER NETWORKS

Teacher: **Phan Trung Phát**

Group: **CS4283.Q12.CTTT**

Name	ID
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OTHER EVALUATIONS (*):

Content	Result
Average total time to complete the lab ⁽¹⁾	10 Hours
Video Link ⁽²⁾ (if any)	
Feedback ⁽³⁾ (if any) + Challenges + Suggestions ...	
Self-Assessment Score ⁽⁴⁾	9/10

(*): Sections (1) and (4) are mandatory.

TASK ASSIGNMENT TABLE:

Members	Task Details
Nguyễn Dương Phúc	Task 1- 13 (Work - 100%)

The section below is the detailed report of the group/individual responsible.



FOR INTERNAL CIRCULATION ONLY



Question 1

1. Evidence:

UDP:

udp				
No.	Time	Source	Destination	Protocol
8334	33.938595	192.168.1.16	192.168.1.10	RTP
8335	33.938595	192.168.1.16	192.168.1.10	RTP
8336	33.938595	192.168.1.16	192.168.1.10	RTP
8337	33.938595	192.168.1.16	192.168.1.10	RTP
8338	33.940278	192.168.1.16	192.168.1.10	RTP
8339	33.940278	192.168.1.16	192.168.1.10	RTP
8340	33.943488	192.168.1.16	192.168.1.10	RTP
8341	33.972070	192.168.1.16	192.168.1.10	RTP
8342	33.972070	192.168.1.16	192.168.1.10	RTP
8343	33.972070	192.168.1.16	192.168.1.10	RTP

TCP:

tcp				
No.	Time	Source	Destination	Protocol
113	1.979886	192.168.1.16	192.168.1.10	TCP
114	1.979909	192.168.1.10	192.168.1.16	TCP
115	1.989762	192.168.1.16	192.168.1.10	TCP
116	1.989762	192.168.1.16	192.168.1.10	TCP
117	1.989826	192.168.1.10	192.168.1.16	TCP
118	2.173511	192.168.1.16	192.168.1.10	TCP
119	2.173511	192.168.1.16	192.168.1.10	TCP
120	2.173511	192.168.1.16	192.168.1.10	TCP
121	2.173662	192.168.1.10	192.168.1.16	TCP

Q6:

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 192.168.1.1
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 70-08-94-D3-CE-87
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:ee0:53a9:28f0:8be0:e83:5a1f:618a(Preferred)
Temporary IPv6 Address. . . . . : 2001:ee0:53a9:28f0:1972:aefa:fd3d:f254(Preferred)
Link-local IPv6 Address . . . . . : fe80::88fe:9b5:cfd:6817%5(Preferred)
IPv4 Address. . . . . : 192.168.1.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, November 9, 2025 1:42:34 PM
Lease Expires . . . . . : Monday, November 10, 2025 1:42:31 PM
Default Gateway . . . . . : fe80::1%5
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 108005524
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-C0-4F-27-C8-53-09-42-00-FA
DNS Servers . . . . . : fe80::1%5
                          8.8.8.8
                          8.4.4.8
                          fe80::1%5
```

2. Explanation:



Question	UDP (Attempt 1 - RTSP Streaming)	TCP (Attempt 2 - HTTP Streaming)
Q1	The video started very fast with almost no delay, but sometimes the image was not smooth, or a few frames were missing.	The video was smooth and clear, but it took a few seconds to start because of buffering.
Q2	Wireshark showed packets with the UDP protocol.	Wireshark showed packets with the TCP protocol.
Q3	The video had low delay (fast response) but sometimes small glitches appeared.	The video had no glitches but there was a short delay before it played.
Q4	The streaming used UDP , which sends data fast but does not check for lost packets.	The streaming used TCP , which checks and resends lost packets for reliable delivery.
Q5	UDP: Faster and better for real-time video but can lose data.	TCP: Slower but more reliable and stable for streaming.

Q6:

IPv4	192.168.1.10
Link-local IPv6 address	fe80::88fe:9b5:cfb:6817
MAC address	70-08-94-D3-CE-87
Default gateway	fe80::1 192.168.1.1
DNS Servers	fe80::1 8.8.8.8, 8.4.4.8

Question 2**1. Evidence:**

dns.qry.name == "uit.edu.vn"					
No.	Time	Source	Destination	Protocol	Length Info
1364	6.033792	10.0.112.86	8.8.8.8	DNS	70 Standard query 0x7c84 HTTPS uit.edu.vn
1365	6.033920	10.0.112.86	8.8.8.8	DNS	70 Standard query 0x0cfc A uit.edu.vn

2. Explanation:

The query for uit.edu.vn is in packet **1365**.

Question 3**1. Evidence:**



```
Domain Name System (Response)
  Transaction ID: 0x0cfc
  ▶ Flags: 0x8180 Standard query response
  Questions: 1
  Answer RRs: 1
  Authority RRs: 0
  Additional RRs: 0
  ▶ Queries
  ▶ Answers
  [Request In: 1365]
  [Time: 0.086482000 seconds]
```

1421	6.117023	8.8.8.8	10.0.112.86	DNS	131	Standard query response 0x7c84 HTTP5 uit.edu.vn SOA ns1.pavietr
1422	6.118450	8.8.8.8	10.0.112.86	DNS	131	Standard query response 0x7c84 HTTP5 uit.edu.vn SOA ns1.pavietr
1423	6.120402	8.8.8.8	10.0.112.86	DNS	86	Standard query response 0x0cfc A uit.edu.vn A 118.69.123.140

2. Explanation:

The response packet to the above query is packet 1423. It can be identified by "Standard query response" in the Info column, and in the Domain Name System section, it shows "Request In: 1365", indicating that it is a response to the query in packet 1365.

Question 4

1. Evidence:

```
udp
```

No.	Time	Source	Destination	Protocol	Length	Info
3	0.304698	10.0.135.111	255.255.255.255	UDP	62	58112 → 34569 Len=20

```
▼ User Datagram Protocol, Src Port: 58112, Dst Port: 34569
  Source Port: 58112
  Destination Port: 34569
  Length: 28
  Checksum: 0x0b36 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Stream Packet Number: 1]
  ▶ [Timestamps]
  UDP payload (20 bytes)
```

2. Explanation:

Packet 3. The fields in the UDP header are:

- Source Port (2 bytes): 16-bit number identifying the port from which the packet is sent.
- Destination Port (2 bytes): 16-bit number identifying the port to which the packet is sent.
- Length (2 bytes): 16-bit number indicating the total length of the UDP packet, including header and data. Minimum length is 8 bytes (header only).
- Checksum (2 bytes): 16-bit number used to verify the integrity of the packet. If the checksum is incorrect, the packet is discarded.



Question 5

1. Evidence:

Source Port:

```
Source Port: 58112  f e3 00 87
```

Destination Port:

```
Destination Port: 34569  0 87 09 00
```

Length:

```
Length: 28  09 00 1c 00
```

Checksum:

```
Checksum: 0x0b36 [unverified]  c 0b 36 ff
```

2. Explanation:

When we click on each field in Wireshark, the corresponding bytes are **highlighted** in the **Hexadecimal view**. **Each** of the four UDP header fields (Source Port, Destination Port, Length, Checksum) occupies **2 bytes**, so 2 hex cells are highlighted for each field.

Question 6

1. Evidence:

```
▼ User Datagram Protocol, Src Port: 58112, Dst Port: 34569
  Source Port: 58112
  Destination Port: 34569
  Length: 28
  Checksum: 0x0b36 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Stream Packet Number: 1]
  ▶ [Timestamps]
  UDP payload (20 bytes)
  ▶ Data (20 bytes)
```

2. Explanation:

The Length field represents the total length of the UDP datagram, including **header** and **data**.

Length = Header (8 bytes) + Data (20 bytes) = 28 bytes, which matches the Length field (28 bytes).

Question 7

1. Evidence:

```

[Stream Packet Number: 1]
  [Timestamps]
    UDP payload (20 bytes)
  Data (20 bytes)

```

2. Explanation:

UDP Payload max = $2^{16} - 1 - 20$ (IP header) - 8 (UDP header) = 65,507 bytes.

In the captured packet, the Data section is 20 bytes, so the UDP payload is 20 bytes, since the payload represents the original data excluding the UDP and IP headers.

Question 8

1. Evidence:

2. Explanation:

The Source Port field in the UDP header is 16 bits long.

A 16-bit field can represent values from 0 to $2^{16} - 1 = 65,535$.

Source Port max = $2^{16} - 1 = 65,535$

Question 9

1. Evidence:

The sent packet

```

...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x985b [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.112.86
Destination Address: 8.8.8.8
[Stream index: 1]

```

The response packet:

```

...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 110
Protocol: UDP (17)
Header Checksum: 0x1e87 [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 10.0.112.86
[Stream index: 1]
User Datagram Protocol Src Port: 53 Dst Port: 61590

```

2. Explanation:



The packet sent by the computer and the response packet have **opposite** Source and Destination IP addresses and ports.

- For the sent packet:
 - Source: 10.0.112.86
 - Destination: 8.8.8.8
- For the response packet:
 - Source: 8.8.8.8
 - Destination: 10.0.112.86

This occurs because each UDP packet contains both a source and a destination IP and port. This ensures that the sender knows where to send the packet and the receiver knows where it came from, allowing communication to reach the correct host and service.

Question 10

1. Evidence:

```
999      10.0.112.86      172.217.25.35      TCP      54 61404 → 443 [ACK] Seq=1 Ack=1 Win=
673      172.217.25.35      10.0.112.86      TCP      66 443 → 61404 [ACK] Seq=1 Ack=1 Win=
▶ Frame 64: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{1935608A-6
▶ Ethernet II, Src: LiteonTechno_d3:ce:87 (70:08:94:d3:ce:87), Dst: Intel_14:2c:d4 (90:e2:ba:14:2c:d4)
▶ Internet Protocol Version 4, Src: 10.0.112.86, Dst: 172.217.25.35
▶ Transmission Control Protocol, Src Port: 61404, Dst Port: 443, Seq: 1772, Ack: 5649, Len: 0
```

2. Explanation:

Client IP: 10.0.112.86

Client TCP port: 61404

Question 11

1. Evidence:

```
47 1.331999      10.0.112.86      172.217.25.35      TCP      54 61404 → 443 [ACK] Se
46 1.331973      172.217.25.35      10.0.112.86      TCP      66 443 → 61404 [ACK] Se
▶ Frame 47: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{1935608A-6
▶ Ethernet II, Src: LiteonTechno_d3:ce:87 (70:08:94:d3:ce:87), Dst: Intel_14:2c:d4 (90:e2:ba:14:2c:d4)
▶ Internet Protocol Version 4, Src: 10.0.112.86, Dst: 172.217.25.35
▶ Transmission Control Protocol, Src Port: 61404, Dst Port: 443, Seq: 1, Ack: 1, Len: 0
```

2. Explanation:

Server IP: 172.217.25.35

Server TCP Port: 443

Question 12

1. Evidence:

```

1000 .... = Header Length: 32 bytes (8)
Flags: 0x012 (SYN, ACK)
 000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... .... 0... = Push: Not set
.... ..... .0.. = Reset: Not set
  .... ...1. = Syn: Set
.... .... ...0 = Fin: Not set
[TCP Flags: .....A..S.]
Window: 65535
[Calculated window size: 65535]
Checksum: 0x6d97 [unverified]

```

2. Explanation:

The TCP SYN segment that initiates the connection between the client and the server has a **SYN flag set (SYN = 1)** and **ACK = 0**. The **Sequence Number** is the client's Initial Sequence Number (ISN), which can be any number, not necessarily 0.

The component that indicates it is a SYN segment is the **Flags field** in the TCP header.

Question 13

1. Evidence:

```

Transmission Control Protocol, Src Port: 443, Dst Port: 61404
Source Port: 443
Destination Port: 61404
[Stream index: 2]
[Stream Packet Number: 2]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1458106221
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 2374761420

```

2. Explanation:

- Sequence Number of the SYN/ACK segment: server's Initial Sequence Number (ISN)
- Acknowledgement Number: client's ISN + 1
- How server determines ACK: by reading the Sequence Number of the received SYN segment from the client



- Component indicating SYN/ACK: the Flags field in the TCP header (SYN = 1, ACK = 1).